

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st March 2017

Subject SA3 – General Insurance

Time allowed: Three Hours (14.45* – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Your property underwriter has requested you to review the “sum insured” data supplied by a large nationalized bank pertaining to its “owned office premises” exposures around the country. Your underwriter wants to support both property damage loss and business interruption loss from these exposures. Your task is to develop a viable pricing plan. As a first step you have planned to estimate the burning cost from these exposures.
- i)** Describe in detail, the steps you would take to estimate the property burning cost expectation. Your answer should include the following:
 - a)** Identify four major variables that would impact property damage losses expected from these exposures. Your answer should clearly explain a measurement proxy for the identified variables and the relationship between the variable and the loss. (4)
 - b)** Identify and explain two special features found in bank branch offices that can add to contents losses. (2)
 - c)** Explain how you would organize the total property exposure data so that the aggregate cost can be justified easily to a non-technical audience. (2)
 - ii)** Describe in detail, the steps you would take to estimate the Business Interruption Loss expectation. Your answer should include the following:
 - a)** Discuss a rational process to classify the different operations of a bank in order to arrive at appropriate exposure figures that can be used to evaluate a bank’s profitability. (4)
 - b)** What primary statistical measures would be needed to characterize the estimate’s properties? Explain how you would be using the statistical measures in your estimation process. (4)
 - c)** Explain how you would organize the total BI exposure data so that the aggregate cost can be justified easily to a non-technical audience. (2)
 - iii)** Identify and explain the characteristics of three additional liability covers that can be added to this basic product to augment total premium collection. Your answer should include the exposure measure, the underlying cost basis, and special features. (5)
 - iv)** Your company’s top management is keen to get this account. Justify the components of a walk away price below which the insurer should not be writing this account. (2)
- [25]**
- Q. 2)** The IRDAI (Assets, Liabilities, and Solvency Margin of General Insurance Business) Regulations, 2016 were issued by IRDAI in 2016 to be effective April 1, 2016 and consequently The IRDA (Assets, Liabilities, and Solvency Margin of Insurance Business) Regulations, 2000 stands superseded.
- i)** List at least six changes brought in the 2016 regulations with respect to valuation of assets and liabilities in General Insurance business that would have an impact on solvency estimation. (4)
 - ii) a)** Define a Premium Deficiency Reserve (PDR). (2)

- b) Explain with rationale the factors/circumstances that could give rise to premium deficiency and corrective measures that could be taken. (4)
- iii) The following information from the financials is made available to you with respect to a new general insurer for the immediately preceding full financial year ended 31st March 2016:

INR Crore

Class of business	NWP	Net UPR	Net Loss Ratio	Net Commission Ratio	Management expense Ratio
Fire	250	150	69%	5%	20%
Health	400	250	75%	10%	20%
Motor	600	350	99%	8%	20%

- a) Using a simple financial year basis, calculate the estimated premium deficiency reserve based on the above information. Set out the assumptions and PDR reserving norms in Indian context. (4)
- b) List the areas in which premium deficiency could be seen as a useful tool in a general insurer's business decision making process. (2)
- iv) Describe, giving reference to applicability to Indian context wherever relevant, other practical considerations with respect to PDR estimation process/methodology that could have an impact on the estimated quantum of premium deficiency reserves. (6)
- [22]

Q. 3) Assume that an optimistic characterization of a typical Indian General Insurer is an annual premium volume in the Rs 5,000 Crore range with a 40, 25, 35 split broadly between premiums from Motor, Fire and Health exposures. Your task is to formulate a stress test for this insurer and instruct the insurer's Board about potential capital needs based on results from the stress test. You can assume that the insurer's current solvency ratio is 2 and that the RSM is Rs 1,500 Crore, and is currently making underwriting profits.

Your response should be in five parts.

- i) **The first three parts** should elaborate the stress situations that are likely to be faced by the insurer in each of the three lines. Your description should include typical numbers, typical situations, typical market and environmental conditions. You will need to split the three broad exposure categories into sub-groups in order to paint a clear picture. Your answer should clearly indicate the underlying causes. Note that there are three parts in this section – one for each line of business. (15)
- ii) **The fourth part** should elaborate on how you would integrate the numerical characteristics developed in the first three parts and arrive at the total stress situation that can lead to additional capital requirements for the insurer. (5)
- iii) **The fifth part** should focus on the development of an adequate protection plan. This section should also include comments on any latent losses that have not been modeled and the opportunities and threats from participating in the Prime Minister's Fasal Bima Yojana. (5)
- [25]

- Q. 4)** A large general insurer in India writes, inter alia, significant business in property line. The insurer has had a combination of reinsurance programmes put in place to protect its exposure. Over the years, despite the treaty results being favourable, the insurer has not been able to see much advantage in its ceding terms.

The Board of the general insurer is keen to know whether the reinsurance programme is efficient and whether it is not only meeting risk objectives but also ensuring shareholder value protection. Therefore, for finalizing treaties of the coming year, the Board has directed that all programmes be evaluated on efficiency and a report be placed alongside the reinsurance programme for approval.

- i)** List the characteristics of an insurer that would have a bearing on the choice of the most effective reinsurance programme. (4)

One of the retail property portfolios of the insurer is the homeowners' contents insurance. The insurer has a steady-sized book that it has been writing for previous ten years. The portfolio is covered under a Quota Share reinsurance programme with 25% ceding and a reinsurance commission of 15%.

The insurer's average gross loss ratio on the book over since inception has been 72% with a standard deviation of 18%. The insurer's expected gross premium income in the coming year is INR 500 Crore.

The Chief Actuary, amongst other methods, is testing the effectiveness of this reinsurance arrangement using the 10-10 testing rule. The test states that the reinsurance program can be termed 'effective' if the reinsurer has a 10% or greater chance of incurring a 10% or greater loss in relation to the present value of ceded premiums under the contract.

- ii) a)** Assuming a lognormal distribution on gross loss ratio, assess the efficiency of the above mentioned Quota Share programme under the 10-10 testing rule. Your answer should include the quantum of reinsurer surplus/deficit as per the above arrangement under 10-10 rule. State the assumptions you would make. (9)
- b)** Would your answer change if the commission was changed to 25% and a loss cap of 80% introduced? Demonstrate and comment. (2)
- c)** Based on the given definition of 10-10 rule, discuss the appropriateness of this methodology for risk transfer testing in general and particularly with respect to the above case. (4)
- iii)** Expanding the terms define the following other measures that are also generally used in the context of reinsurance risk transfer tests:
- a)** ERD
- b)** CTE (4)
- iv)** What are the important criteria and considerations that need to be pre-evaluated before putting a reinsurance arrangement through a testing process, like in the above case? (5)

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